

Question				Marking details	Marks Available					
					AO1	AO2	AO3	Total	Maths	Prac
7	(a)			Windows at 15% 4 correct 3 marks Floor at 10% 2 or 3 correct 2 marks Roof at 25% 1 correct 1 mark Walls at 35% 0 correct 0 mark		3		3	1	
	(b)	(i)		Fibre-glass is an insulator or it contains air or it contains air pockets or <u>reduces</u> conduction through the ceiling (1) Don't accept traps air <u>Reduces</u> convection <u>currents</u> in the loft or air in the loft heats up less (1)	2			2		
		(ii)		3 × (1): Ticks in boxes alongside statements 2, 4 and 5 i.e. The required thickness of loft insulation in 2000 is 8 times thicker than in 1970. A house built in 1980 needs 210 mm of loft insulation added to bring it up to 2015 standards. The general trend of the graph indicates that the thickness of required loft insulation has increased at an increasing rate. Deduct 1 mark for each additional tick		3		3	3	
		(iii)		$4.50 \times 120 = \text{£}540$ (1) $\frac{540}{98} = 5.51$ or 5.5 or 6 [years] (1) Don't accept 5 [years] Answer of 0.0459 [years] award 1 mark only		2		2	2	
		(iv)		(1) for either calculation: Insulation 1 saving $(40 - 5.0) \times \text{£}84 = \text{£}2\,940$ Insulation 2 saving $(40 - 6.0) \times \text{£}111 = \text{£}3\,774$ (1) for second calculation and comment/agreement with builder Alternative 1: (1) for either calculation: $5 \times 84 = \text{£}420$ and $40 \times 84 = \text{£}3\,360$ and the difference = $\text{£}2\,940$ $6 \times 111 = \text{£}666$ and $40 \times 111 = \text{£}4\,440$ and the difference = $\text{£}3\,774$ (1) for second calculation and comment/agreement with builder			2	2	1	

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				Alternative 2: $5 \times 84 = \text{£}420$ and $6 \times 111 = \text{£}666$ or $40 \times 84 = \text{£}3\,360$ and $40 \times 111 = \text{£}4\,440$ (1) The differences are $\text{£}3\,774$ and $\text{£}2\,940$ so the builder is correct (1) Alternative 3: (1) for either calculation: $3.5 \times 120 = \text{£}420$ and $40 \times 84 = \text{£}3\,360$ and the difference = $\text{£}2\,940$ $5.5 \times 120 = \text{£}666$ and $40 \times 111 = \text{£}4\,440$ and the difference = $\text{£}3\,774$ (1) for second calculation and comment/agreement with builder Alternative 4: $3.5 \times 120 = \text{£}420$ and $5.5 \times 120 = \text{£}666$ or $40 \times 84 = \text{£}3\,360$ and $40 \times 111 = \text{£}4\,440$ (1) The differences are $\text{£}3\,774$ and $\text{£}2\,940$ so the builder is correct (1) Alternative 5: (1) for either calculation: $(111 - 84) \times 40 = \text{£}1\,080$ or $666 - 420 = \text{£}246$ $\text{£}1\,080 - \text{£}246 = \text{£}834$ (1) and comment/agreement with builder N.B. Any reference to insulation 3 treat as neutral						
				Question 7 total	2	8	2	12	7	0